SG No. 68 D_{2h}^{22} *Ccce* [orthorhombic]

* plus set: $+[0,\ 0,\ 0]$, $+\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$
* Wyckoff site: `4a`, site symmetry: `222`

Table 1: Wyckoff bond: `4a@4a`, bond symmetry: `222`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$\left[0,\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,2,-3,-4]`
2	$[0,\ 0,\ -Z]$	$\left[0,\ \frac{3}{4},\ \frac{3}{4}\right]$	`[5,6,-7,-8]`

Table 2: Wyckoff bond: `4b@4a`, bond symmetry: `222`

No.	vector	center	mapping
1	$[0,\ Y,\ 0]$	$\left[0,\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,-2,3,-4]`
2	$[0,\ -Y,\ 0]$	$\left[0,\ \frac{3}{4},\ \frac{3}{4}\right]$	`[5,-6,7,-8]`

Table 3: Wyckoff bond: `4c@4a`, bond symmetry: `222`

No.	vector	center	mapping
1	$[X,\ 0,\ 0]$	$\left[0,\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,-2,-3,4]`
2	$[-X,\ 0,\ 0]$	$\left[0,\ \frac{3}{4},\ \frac{3}{4}\right]$	`[5,-6,-7,8]`

Table 4: Wyckoff bond: `8d@4a`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\ Y,\ 0]$	$\left[0,\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,-2]`
2	$[-X,\ Y,\ 0]$	$\left[0,\ \frac{1}{4},\ \frac{1}{4}\right]$	`[3,-4]`
3	$[-X,\ -Y,\ 0]$	$\left[0,\ \frac{3}{4},\ \frac{3}{4}\right]$	`[5,-6]`
4	$[X,\ -Y,\ 0]$	$\left[0,\ \frac{3}{4},\ \frac{3}{4}\right]$	`[7,-8]`

Table 5: Wyckoff bond: `8e@4a`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X,\ 0,\ Z]$	$\left[0,\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,-3]`
2	$[-X,\ 0,\ Z]$	$\left[0,\ \frac{1}{4},\ \frac{1}{4}\right]$	`[2,-4]`
3	$[-X,\ 0,\ -Z]$	$\left[0,\ \frac{3}{4},\ \frac{3}{4}\right]$	`[5,-7]`
4	$[X,\ 0,\ -Z]$	$\left[0,\ \frac{3}{4},\ \frac{3}{4}\right]$	`[6,-8]`

Table 6: Wyckoff bond: `8f@4a`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0,\ Y,\ Z]$	$\left[0,\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,-4]`
2	$[0,\ -Y,\ Z]$	$\left[0,\ \frac{1}{4},\ \frac{1}{4}\right]$	`[2,-3]`
3	$[0,\ -Y,\ -Z]$	$\left[0,\ \frac{3}{4},\ \frac{3}{4}\right]$	`[5,-8]`

continued ...

Table 6

No.	vector	center	mapping
4	$[0, Y, -Z]$	$\left[0, \frac{3}{4}, \frac{3}{4}\right]$	[6,-7]

Table 7: Wyckoff bond: 16g@4a, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, \frac{1}{4}, \frac{1}{4}\right]$	[1]
2	$[-X, -Y, Z]$	$\left[0, \frac{1}{4}, \frac{1}{4}\right]$	[2]
3	$[-X, Y, -Z]$	$\left[0, \frac{1}{4}, \frac{1}{4}\right]$	[3]
4	$[X, -Y, -Z]$	$\left[0, \frac{1}{4}, \frac{1}{4}\right]$	[4]
5	$[-X, -Y, -Z]$	$\left[0, \frac{3}{4}, \frac{3}{4}\right]$	[5]
6	$[X, Y, -Z]$	$\left[0, \frac{3}{4}, \frac{3}{4}\right]$	[6]
7	$[X, -Y, Z]$	$\left[0, \frac{3}{4}, \frac{3}{4}\right]$	[7]
8	$[-X, Y, Z]$	$\left[0, \frac{3}{4}, \frac{3}{4}\right]$	[8]

* Wyckoff site: 4b, site symmetry: 222

Table 8: Wyckoff bond: 4a@4b, bond symmetry: 222

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	[1,2,-3,-4]
2	$[0, 0, -Z]$	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	[5,6,-7,-8]

Table 9: Wyckoff bond: 4b@4b, bond symmetry: 222

No.	vector	center	mapping
1	$[0, Y, 0]$	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	[1,-2,3,-4]
2	$[0, -Y, 0]$	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	[5,-6,7,-8]

Table 10: Wyckoff bond: 4c@4b, bond symmetry: 222

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	[1,-2,-3,4]
2	$[-X, 0, 0]$	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	[5,-6,-7,8]

Table 11: Wyckoff bond: 8d@4b, bond symmetry: ..2

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	[1,-2]
2	$[-X, Y, 0]$	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	[3,-4]
3	$[-X, -Y, 0]$	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	[5,-6]
4	$[X, -Y, 0]$	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	[7,-8]

Table 12: Wyckoff bond: `8e@4b`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	`[1,-3]`
2	$[-X, 0, Z]$	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	`[2,-4]`
3	$[-X, 0, -Z]$	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	`[5,-7]`
4	$[X, 0, -Z]$	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	`[6,-8]`

Table 13: Wyckoff bond: `8f@4b`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0, Y, Z]$	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	`[1,-4]`
2	$[0, -Y, Z]$	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	`[2,-3]`
3	$[0, -Y, -Z]$	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	`[5,-8]`
4	$[0, Y, -Z]$	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	`[6,-7]`

Table 14: Wyckoff bond: `16g@4b`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	`[3]`
4	$[X, -Y, -Z]$	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	`[4]`
5	$[-X, -Y, -Z]$	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	`[5]`
6	$[X, Y, -Z]$	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	`[6]`
7	$[X, -Y, Z]$	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	`[7]`
8	$[-X, Y, Z]$	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	`[8]`

* Wyckoff site: `8c`, site symmetry: `-1`

Table 15: Wyckoff bond: `8a@8c`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[1,-5]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	`[2,-6]`
3	$[-X, Y, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[3,-7]`
4	$[X, -Y, -Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[4,-8]`

* Wyckoff site: `8d`, site symmetry: `-1`

Table 16: Wyckoff bond: `8a@8d`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, 0]$	`[1,-5]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[2,-6]`
3	$[-X, Y, -Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[3,-7]`
4	$[X, -Y, -Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[4,-8]`

* Wyckoff site: `8e`, site symmetry: `2..`

Table 17: Wyckoff bond: `8a@8e`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0,\, Y,\, Z]$	$\left[x,\, \frac{1}{4},\, \frac{1}{4}\right]$	`[1,-4]`
2	$[0,\, -Y,\, Z]$	$\left[\frac{1}{2}-x,\, \frac{3}{4},\, \frac{1}{4}\right]$	`[2,-3]`
3	$[0,\, -Y,\, -Z]$	$\left[-x,\, \frac{3}{4},\, \frac{3}{4}\right]$	`[5,-8]`
4	$[0,\, Y,\, -Z]$	$\left[x+\frac{1}{2},\, \frac{1}{4},\, \frac{3}{4}\right]$	`[6,-7]`

Table 18: Wyckoff bond: `8b@8e`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X,\, 0,\, 0]$	$\left[x,\, \frac{1}{4},\, \frac{1}{4}\right]$	`[1,4]`
2	$[-X,\, 0,\, 0]$	$\left[\frac{1}{2}-x,\, \frac{3}{4},\, \frac{1}{4}\right]$	`[2,3]`
3	$[-X,\, 0,\, 0]$	$\left[-x,\, \frac{3}{4},\, \frac{3}{4}\right]$	`[5,8]`
4	$[X,\, 0,\, 0]$	$\left[x+\frac{1}{2},\, \frac{1}{4},\, \frac{3}{4}\right]$	`[6,7]`

Table 19: Wyckoff bond: `16c@8e`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\, Y,\, Z]$	$\left[x,\, \frac{1}{4},\, \frac{1}{4}\right]$	`[1]`
2	$[-X,\, -Y,\, Z]$	$\left[\frac{1}{2}-x,\, \frac{3}{4},\, \frac{1}{4}\right]$	`[2]`
3	$[-X,\, Y,\, -Z]$	$\left[\frac{1}{2}-x,\, \frac{3}{4},\, \frac{1}{4}\right]$	`[3]`
4	$[X,\, -Y,\, -Z]$	$\left[x,\, \frac{1}{4},\, \frac{1}{4}\right]$	`[4]`
5	$[-X,\, -Y,\, -Z]$	$\left[-x,\, \frac{3}{4},\, \frac{3}{4}\right]$	`[5]`
6	$[X,\, Y,\, -Z]$	$\left[x+\frac{1}{2},\, \frac{1}{4},\, \frac{3}{4}\right]$	`[6]`
7	$[X,\, -Y,\, Z]$	$\left[x+\frac{1}{2},\, \frac{1}{4},\, \frac{3}{4}\right]$	`[7]`
8	$[-X,\, Y,\, Z]$	$\left[-x,\, \frac{3}{4},\, \frac{3}{4}\right]$	`[8]`

* Wyckoff site: `8f`, site symmetry: `.2.`

Table 20: Wyckoff bond: `8a@8f`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X,\, 0,\, Z]$	$\left[0,\, y,\, \frac{1}{4}\right]$	`[1,-3]`
2	$[-X,\, 0,\, Z]$	$\left[\frac{1}{2},\, -y,\, \frac{1}{4}\right]$	`[2,-4]`
3	$[-X,\, 0,\, -Z]$	$\left[0,\, -y,\, \frac{3}{4}\right]$	`[5,-7]`
4	$[X,\, 0,\, -Z]$	$\left[\frac{1}{2},\, y,\, \frac{3}{4}\right]$	`[6,-8]`

Table 21: Wyckoff bond: `8b@8f`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[0,\, Y,\, 0]$	$\left[0,\, y,\, \frac{1}{4}\right]$	`[1,3]`
2	$[0,\, -Y,\, 0]$	$\left[\frac{1}{2},\, -y,\, \frac{1}{4}\right]$	`[2,4]`
3	$[0,\, -Y,\, 0]$	$\left[0,\, -y,\, \frac{3}{4}\right]$	`[5,7]`

4	$[0, Y, 0]$	$\left[\frac{1}{2}, y, \frac{3}{4}\right]$	`[6,8]`

Table 22: Wyckoff bond: `16c@8f`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, y, \frac{1}{4}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}, -y, \frac{1}{4}\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[0, y, \frac{1}{4}\right]$	`[3]`
4	$[X, -Y, -Z]$	$\left[\frac{1}{2}, -y, \frac{1}{4}\right]$	`[4]`
5	$[-X, -Y, -Z]$	$\left[0, -y, \frac{3}{4}\right]$	`[5]`
6	$[X, Y, -Z]$	$\left[\frac{1}{2}, y, \frac{3}{4}\right]$	`[6]`
7	$[X, -Y, Z]$	$\left[0, -y, \frac{3}{4}\right]$	`[7]`
8	$[-X, Y, Z]$	$\left[\frac{1}{2}, y, \frac{3}{4}\right]$	`[8]`

* Wyckoff site: `8g`, site symmetry: `..2`

Table 23: Wyckoff bond: `8a@8g`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[0, \frac{1}{4}, z\right]$	`[1,-2]`
2	$[-X, Y, 0]$	$\left[0, \frac{1}{4}, \frac{1}{2} - z\right]$	`[3,-4]`
3	$[-X, -Y, 0]$	$\left[0, \frac{3}{4}, -z\right]$	`[5,-6]`
4	$[X, -Y, 0]$	$\left[0, \frac{3}{4}, z + \frac{1}{2}\right]$	`[7,-8]`

Table 24: Wyckoff bond: `8b@8g`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[0, \frac{1}{4}, z\right]$	`[1,2]`
2	$[0, 0, -Z]$	$\left[0, \frac{1}{4}, \frac{1}{2} - z\right]$	`[3,4]`
3	$[0, 0, -Z]$	$\left[0, \frac{3}{4}, -z\right]$	`[5,6]`
4	$[0, 0, Z]$	$\left[0, \frac{3}{4}, z + \frac{1}{2}\right]$	`[7,8]`

Table 25: Wyckoff bond: `16c@8g`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, \frac{1}{4}, z\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[0, \frac{1}{4}, z\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[0, \frac{1}{4}, \frac{1}{2} - z\right]$	`[3]`
4	$[X, -Y, -Z]$	$\left[0, \frac{1}{4}, \frac{1}{2} - z\right]$	`[4]`
5	$[-X, -Y, -Z]$	$\left[0, \frac{3}{4}, -z\right]$	`[5]`
6	$[X, Y, -Z]$	$\left[0, \frac{3}{4}, -z\right]$	`[6]`
7	$[X, -Y, Z]$	$\left[0, \frac{3}{4}, z + \frac{1}{2}\right]$	`[7]`
8	$[-X, Y, Z]$	$\left[0, \frac{3}{4}, z + \frac{1}{2}\right]$	`[8]`

* Wyckoff site: `8h`, site symmetry: `..2`

Table 26: Wyckoff bond: `8a@8h`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[\frac{1}{4}, 0, z\right]$	`[1,-2]`
2	$[-X, Y, 0]$	$\left[\frac{3}{4}, 0, \frac{1}{2} - z\right]$	`[3,-4]`
3	$[-X, -Y, 0]$	$\left[\frac{3}{4}, 0, -z\right]$	`[5,-6]`
4	$[X, -Y, 0]$	$\left[\frac{1}{4}, 0, z + \frac{1}{2}\right]$	`[7,-8]`

Table 27: Wyckoff bond: `8b@8h`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[\frac{1}{4}, 0, z\right]$	`[1,2]`
2	$[0, 0, -Z]$	$\left[\frac{3}{4}, 0, \frac{1}{2} - z\right]$	`[3,4]`
3	$[0, 0, -Z]$	$\left[\frac{3}{4}, 0, -z\right]$	`[5,6]`
4	$[0, 0, Z]$	$\left[\frac{1}{4}, 0, z + \frac{1}{2}\right]$	`[7,8]`

Table 28: Wyckoff bond: `16c@8h`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{4}, 0, z\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{4}, 0, z\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[\frac{3}{4}, 0, \frac{1}{2} - z\right]$	`[3]`
4	$[X, -Y, -Z]$	$\left[\frac{3}{4}, 0, \frac{1}{2} - z\right]$	`[4]`
5	$[-X, -Y, -Z]$	$\left[\frac{3}{4}, 0, -z\right]$	`[5]`
6	$[X, Y, -Z]$	$\left[\frac{3}{4}, 0, -z\right]$	`[6]`
7	$[X, -Y, Z]$	$\left[\frac{1}{4}, 0, z + \frac{1}{2}\right]$	`[7]`
8	$[-X, Y, Z]$	$\left[\frac{1}{4}, 0, z + \frac{1}{2}\right]$	`[8]`

* Wyckoff site: `16i`, site symmetry: `1`

Table 29: Wyckoff bond: `16a@16i`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, z]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2} - x, -y, z\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[-x, y, \frac{1}{2} - z\right]$	`[3]`
4	$[X, -Y, -Z]$	$\left[x + \frac{1}{2}, -y, \frac{1}{2} - z\right]$	`[4]`
5	$[-X, -Y, -Z]$	$[-x, -y, -z]$	`[5]`
6	$[X, Y, -Z]$	$\left[x + \frac{1}{2}, y, -z\right]$	`[6]`
7	$[X, -Y, Z]$	$\left[x, -y, z + \frac{1}{2}\right]$	`[7]`
8	$[-X, Y, Z]$	$\left[\frac{1}{2} - x, y, z + \frac{1}{2}\right]$	`[8]`